

Permutations and Combinations

→ Permutations:- it means arrangement of things, suppose there are n different objects, we can arrange ' r ' of these objects. It can be done in ~~nPr~~ nPr ways.

$$P(n, r) = {}^n P_r = \frac{n!}{(n-r)!}$$

→ Permutation with repetition:- suppose it is required to find the number of permutation that can be formed from the collection of N objects of which n_1 are of type I and n_2 are of type II and n_k are of type K.

∴ Total no. of permutations is given by -

$$\frac{n!}{n_1! n_2! \dots n_k!}$$

→ Combination:- it means selection of things, suppose there are n different objects and we wish to select ' r ' of these objects. It can be done in ${}^n C_r$ ways.

$$C(n, r) = {}^n C_r = \frac{n!}{(n-r)! r!}$$

→ Combination with repetition:- the total number of possible ways of distributing r identical objects from n different objects is known as combination with repetition and is given by -

$${}^{n+r-1} C_r = \frac{(n+r-1)!}{(n-1)! r!}$$

① Example

$${}_5 P_2 =$$

② Example:

Q1:- How many can be selected

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Q2:- Find
word
4 dist

① Example :- 5 - ABCDE

$${}^5P_2 = \frac{5!}{(5-2)!} = \frac{5!}{3!} = \frac{5 \times 4 \times 3!}{3!} = \underline{\underline{20}}$$

② Example :- ABC

BCA
CBA
ACB
BAC
CAB

$\therefore n = 3$
 $n = 3$

$$\therefore {}^3C_3 = \frac{3!}{(3-3)! \cdot 3!} = \frac{1}{0!}$$

Q1:- How many committees of 5 with given chair person can be selected from 12 persons.

Solⁿ - The chair person can be selected in ${}^{12}C_1$ ways.
From 11 person we need to form a committee 4 members can be chosen in ${}^{11}C_4$ ways
 $\therefore$ the possible number of such committee is ${}^{12}C_1 \times {}^{11}C_4$

$$\begin{aligned} {}^{12}C_1 \times {}^{11}C_4 &= \frac{12!}{(12-1)! \cdot 1!} \times \frac{11!}{(11-4)! \cdot 4!} \\ &= \frac{12!}{11!} \times \frac{11!}{7! \cdot 4!} = \frac{12 \times 11 \times 10 \times 9 \times 8 \times 7}{4 \times 3 \times 2 \times 1} \\ &= \underline{\underline{3960}} \end{aligned}$$

Q2:- Find the number of permutations of the letter of the word CHEMISTRY. Also in how many way words of 4 distinct letter can be form.

solⁿ:- $n = 9, r = 9$

$${}^n P_r = \frac{n!}{(n-r)!} = \frac{9!}{(9-9)!} = \frac{9!}{0!} = 9$$

$$= \frac{9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2}{1}$$

$$= \underline{\underline{362880}}$$

$n = 9, r = 4$

$${}^n P_r = \frac{n!}{(n-r)!} = \frac{9!}{(9-4)!} = \frac{9!}{5!}$$

$$= \frac{9 \times 8 \times 7 \times 6 \times \cancel{5} \times 4 \times 3 \times 2}{\cancel{5} \times 4 \times 3 \times 2}$$

$$= \underline{\underline{3024}}$$

Q3:- Find the number of permutations of the letters of the following words.

(i) PROGRESS

$$\frac{n!}{n_1! n_2! \dots n_k!}$$

$R = n_1$

$S = n_2$

$n = 8$

$$= \frac{8!}{2! \times 2!}$$

$$= \frac{8 \times 7 \times \cancel{6} \times 5 \times 4 \times 3 \times \cancel{2}}{2 \times 1 \times 2 \times 1} = \underline{\underline{10080}}$$

(ii) MATHEMATICS

~~M~~ = $n_1, M = n_1 = 2$

$A = n_2 = 2$

$T = n_3 = 2$

$$= \frac{11!}{2! 2! 2! 2! 2!}$$

$$= \frac{11 \times 10 \times 9 \times 8 \times 7 \times 6!}{2 \times 1 \times 2 \times 1 \times 2 \times 1}$$

$$= \underline{\underline{4989600}}$$

(ii) ENGINEERING.

$$E = n_1 = 3$$

$$N = n_2 = 3$$

$$G = n_3 = 2$$

$$I = n_4 = 2$$

$$= \frac{11!}{3! 3! 2! 2!}$$

$$= \frac{11 \times 10 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}{3 \times 2 \times 1 \times 3 \times 2 \times 1 \times 2 \times 1 \times 2 \times 1}$$

$$= \underline{\underline{277200}}$$

(10) DIFFICULT

$$n = 9$$

$$I = n_1 = 2$$

$$F = n_2 = 2$$

$$= \frac{9!}{2! 2! 2!}$$

$$= \frac{9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2}{2 \times 1 \times 2 \times 1 \times 2 \times 1}$$

$$= \underline{\underline{90720}}$$

Q4: How many different streams of length 4 can be formed using the letters of the word FLOWER

$$n = 6, r = 4$$

$${}^n P_r = \frac{n!}{(n-r)!}$$

$$= \frac{6!}{(6-4)!} = \frac{6!}{2!}$$

$$= \frac{6 \times 5 \times 4 \times 3 \times 2}{2 \times 1} = 360$$

Q5:- A certain question paper contains 2 parts part A and part B each containing 4 questions, how many different ways a student can answer 5 questions by selecting atleast 2 questions from each part.

	Part A	Part B	
	4	4	
Case I >	Part A	Part B	
	3	2	
Case I >	Part A	Part B	
	4	4	

$${}^4C_3 \times {}^4C_2 = 24$$

	Part A	Part B	
Case II >	4	1	
	4C_2	$\times {}^4C_3$	$= 24$

$\therefore$ the student can answer the question paper either by case I or case II, this can be done in $24 + 24 = 48$ ways.

Q6:- Find the value of n if ${}^np_2 = 132$

$$\begin{aligned} {}^np_2 &= \frac{n!}{(n-2)!} = 132 \\ &= \frac{n(n-1)(n-2)!}{(n-2)!} = 132 \\ &= n^2 - n = 132 \end{aligned}$$

$$\Rightarrow n^2 - n - 132 = 0$$

$$\Rightarrow 11^2 - 12 - 132 = 0$$

$$\Rightarrow n(n-2) + 1(n-12) = 0$$

$$\Rightarrow (n-12)(n+11) = 0$$

$$= n = 12$$

$$2) {}^n P_2 = 3 \times {}^n P_2$$

$$\frac{n!}{(n-2)!} = 3 \times \frac{n!}{(n-2)!}$$

$$= \frac{n(n-1)(n-2)(n-3)}{n-3} = \cancel{3 \times n(n-1)(n-2)}$$

$$\frac{3 \times n(n-1)(n-2)!}{(n-2)!}$$

$$= \frac{n(n-1)(n-2)(n-3)}{(n-3)} = \frac{3 \times n(n-1)(n-2)!}{(n-2)!}$$

$$= n(n^2 - 2n - n + 2) = 3 \times n^2 - n$$

$$= n(n^2 - 3n + 2) = 3n^2 - n$$

$$= n^3 - 3n^2 + 2n = 3n^2 - n$$

$$\Rightarrow n^3 + 2n + n = 0$$

$$\Rightarrow n^3 + 3n = 0$$

$$n - 2 = 3$$

$$\Rightarrow n = 5$$

$$3) {}^n P_5 = 20 \times {}^n P_3$$

$$\frac{n!}{(n-5)!} = 20 \times \frac{n!}{(n-3)!}$$

$$\Rightarrow \frac{n(n-1)(n-2)(n-3)(n-4)(n-5)!}{(n-5)!} = \frac{20 \times n(n-1)(n-2)(n-3)!}{(n-3)!}$$

$$\Rightarrow n(n-3)(n-4) = 20n$$

$$\Rightarrow n(n^2 - 4n - 3n + 12) = 20n$$

$$\Rightarrow n(n^2 - 7n + 12) = 20n$$

$$\Rightarrow n^3 - 7n^2 + 12n = 20n$$

$$\Rightarrow n^3 - 8n^2 + 12n - 8 = 0$$

$$\Rightarrow n^2(n-8) + (n-8) = 0$$

$$\Rightarrow (n+1)(n-8) = 0$$

$$n+120$$

$$n = -120 (x)$$

$$n-8=0$$

$$n=8 (v)$$

$$4) \quad {}^n P_5 = 42 \times {}^n P_3$$

$$\frac{n(n-1)(n-2)(n-3)(n-4)(n-5)!}{(n-5)!} = 42 \times \frac{n(n-1)(n-2)(n-3)!}{(n-3)!}$$

$$\Rightarrow (n-3)(n-4) = 42n$$

$$\Rightarrow n^2 - 4n - 3n + 12 = 42$$

$$\Rightarrow n^2 - 7n + 12 - 42 = 0$$

$$\Rightarrow n^2 - 7n - 30 = 0$$

$$\Rightarrow n^2 - 10n + 3n - 30 = 0$$

$$\Rightarrow n(n-10) + 3(n-10) = 0$$

$$\Rightarrow (n-10)(n+3) = 0$$

$$\Rightarrow \underline{n=10}$$

$$5) \quad \frac{{}^n P_4}{{}^{n-1} P_4} = \frac{5}{3}$$

$$\frac{\frac{n!}{(n-4)!}}{\frac{(n-1)!}{(n-1-4)!}} = \frac{5}{3}$$

$$= \frac{n!}{(n-4)!} \times \frac{(n-5)!}{(n-1)!} = \frac{5}{3}$$

$$\Rightarrow \frac{n(n-1)(n-2)(n-3)(n-4)!}{(n-4)!} \times \frac{(n-5)!}{(n-1)(n-2)(n-3)(n-4)!}$$

$$\Rightarrow \frac{n}{n-4} = \frac{5}{3}$$

$$\Rightarrow 3n = 5n - 20$$

$$\Rightarrow 5n - 3n = 20$$

$$\Rightarrow 2n = 20 \Rightarrow \underline{n=10}$$

6) 6 boys and 4 girls got elected as CRs in a college a 5 members student council has to be formed from the elected class representatives. In how many ways this council can be formed such that there are

- (i) 3 boys and 2 girls
- (ii) Atleast 2 girls
- (iii) Atleast 3 boys
- (iv) Atmost 2 boys
- (v) Atmost 2 girls

(i) 6 boys 4 girls

$$(i) {}^6C_3 \times {}^4C_2$$

$${}^nC_r = \frac{n!}{(n-r)!r!}$$

$${}^6C_3 = \frac{6!}{(6-3)!3!} \times \frac{4!}{(4-2)!2!}$$

$$= \frac{6 \times 5 \times 4 \times 3 \times 2 \times 1}{3 \times 2 \times 3 \times 2} \times \frac{4 \times 3 \times 2 \times 1}{2 \times 2}$$

$$= 120 \text{ ways.}$$

- (ii) Case 1 $\rightarrow$ 3 boys and 2 girls $\rightarrow {}^6C_3 \times {}^4C_2$
 Case 2 $\rightarrow$ 2 boys and 3 girls $\rightarrow {}^6C_2 \times {}^4C_3$
 Case 3 $\rightarrow$ 1 boy and 4 girls $\rightarrow {}^6C_1 \times {}^4C_4$

$$\text{Case 1} \rightarrow \frac{6!}{(6-3)!3!} \times \frac{4!}{(4-2)!2!}$$

$$= \frac{6 \times 5 \times 4 \times 3 \times 2 \times 1}{3 \times 2 \times 3 \times 2} = \frac{4 \times 3 \times 2 \times 1}{2 \times 2}$$

$$= 120 \text{ ways}$$

$$\text{Case 2} \quad \frac{6!}{(6-2)!2!} \times \frac{4!}{(4-3)!3!}$$

$$= \frac{6 \times 5 \times \cancel{4} \times \cancel{3} \times \cancel{2}}{\cancel{4} \times \cancel{3} \times \cancel{2}} \times \frac{\cancel{4}^2 \times \cancel{3} \times \cancel{2}}{1 \times \cancel{3} \times \cancel{2}}$$

$$= 30 \times 2$$

$$= 60$$

$$\therefore 120 + 60 = 180 \text{ ways}$$

$$\text{Case 3} \quad \frac{6!}{(6-1)!1!} \times \frac{4!}{(4-4)!4!}$$

$$\frac{6 \times \cancel{5} \times \cancel{4} \times \cancel{3} \times \cancel{2}}{\cancel{5} \times \cancel{4} \times \cancel{3} \times \cancel{2} \times 1} \times \frac{\cancel{4} \times \cancel{3} \times \cancel{2}}{\cancel{4} \times \cancel{3} \times \cancel{2}}$$

$$\therefore 120 + 60 + 6 = 186 \text{ ways}$$

(ii) Case 1 $\rightarrow$ 3 boys and 2 girls $\rightarrow {}^6C_3 \times {}^4C_2 = 120$

Case 2 $\rightarrow$ 4 boys and 1 girl $\rightarrow {}^6C_4 \times {}^4C_1 = 60$

Case 3 $\rightarrow$ 5 boys and 0 girls $\rightarrow {}^6C_5 \times {}^4C_0 = 6$

(iv) Case 1 $\rightarrow$ 1 boy and 4 girls
 ${}^6C_1 \times {}^4C_4 = 6$

Case 2 $\rightarrow$ 2 boys and 3 girls
 ${}^6C_2 \times {}^4C_3 = 60 + 6 = 66$

(v)

- 7) Find the number of permutations of the letters of the word ENGINEERING such that all
- the E's are together.
 - G and R are next to each other
 - All the vowels are adjacent
 - Arrangement begins with N.

ENGINEERING

11!

$$\begin{aligned}
 & \frac{3! \cdot 3! \cdot 2! \cdot 2!}{11 \times 10 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1} \\
 & \frac{3 \times 2 \times 1 \times 3 \times 2 \times 1 \times 2 \times 1 \times 2 \times 1}{277200} \\
 & = 277200
 \end{aligned}$$

(i) (EEE) $\overbrace{\text{NNNGIIR}}^8$

$$n = 8 + 1 = 9$$

$$\begin{aligned}
 & \frac{9!}{1! \cdot 3! \cdot 2! \cdot 2! \cdot 1!} = \frac{9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2}{1 \times 3 \times 2 \times 1 \times 2 \times 1 \times 2 \times 1 \times 1} \\
 & = 15120
 \end{aligned}$$

$$EEE \Rightarrow n = 3$$

$$\frac{3!}{3} = 1$$

$$\therefore 15120 \times 1 = 15120$$

(ii) (GGR) $\overbrace{\text{EEENNII}}^8$

$$n = 9$$

$$\begin{aligned}
 & \frac{9!}{1! \cdot 3! \cdot 3! \cdot 2!} = \frac{9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2}{3 \times 2 \times 3 \times 2 \times 2 \times 1} \\
 & = 5040
 \end{aligned}$$

$$GGR \Rightarrow n=3; n_1=G=2 \Rightarrow \frac{3!}{2!1!}$$

$$n_2=R=1$$

$$\therefore 5040 \times 3 = 15120$$

$$(iii) \underset{1}{(EEEEII)} \underset{6}{GGNNNR}$$

$$n=7$$

$$\frac{7!}{1!2!3!} = \frac{7 \times \cancel{6} \times 5 \times 4 \times \cancel{3} \times \cancel{2}}{\cancel{3} \times \cancel{2} \times \cancel{2}} = \underline{\underline{420}}$$

$$EEEEII \Rightarrow n=5; n_1=3$$

$$\frac{5!}{3!2!} = \frac{5 \times \cancel{4} \times \cancel{3} \times \cancel{2}}{\cancel{3} \times \cancel{2} \times \cancel{2}} = \underline{\underline{10}}$$

$$(iv) \underset{10}{(NN)EEEE NN} \underset{10}{GGIIR}$$

$$\frac{n=10}{10} = \frac{10 \times 9 \times 8 \times 7 \times \cancel{6} \times 5 \times \cancel{4} \times \cancel{3} \times \cancel{2}}{\cancel{3} \times \cancel{2} \times \cancel{2} \times 1 \times \cancel{7} \times \cancel{2} \times 1 \times 1} = \frac{280 \times 7 \times 3}{10 \times 10 \times 10} = \underline{\underline{5880}}$$

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Q1:- If

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